GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of canopy height compared to reference sites

MB

2026-06-02

Contents

> **NOTE:** To compare different models, you only have to change the models in the section 'Load models'

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```r
library(here)
library(tidyverse)
```

Packages

```
## Warning: Paket 'ggplot2' wurde unter R Version 4.5.3 erstellt
```

```
## Warning: Paket 'purrr' wurde unter R Version 4.5.3 erstellt
```

```
## Warning: Paket 'dplyr' wurde unter R Version 4.5.3 erstellt
```

```
library(ggbeeswarm)
library(patchwork)
library(DHARMa)
library(emmeans)
```

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites_refs.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
      ),
    obs.year = "f",
    hydrology = col_factor(levels = c("moist", "fresh", "dry"), ordered = TRUE),
    mngm.type = "f"
  )
) %>%
  rename(y = cwm.abu.height) #%>%
  #filter(y < 1) # see section Outliers: Exclude site N_DAM (more or less only the tall grass Arrhenath
```

Load data

Statistics

Data exploration

Means and deviations

```
Rmisc::CI(sites$y, ci = .95)
```

```
##     upper      mean     lower
## 0.4842538 0.4730696 0.4618855
```

```
median(sites$y)
```

```
## [1] 0.47
```

```
sd(sites$y)
```

```
## [1] 0.143179
```

```
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##   5%  95%
## 0.27 0.70
```

```
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id     n
##   <fct>  <int>
```

```
## 1 654      209
## 2 664      212
## 3 686      211
```

```r
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type     n
##   <ord>     <int>
## 1 positive     57
## 2 restored    472
## 3 negative    103
```

```r
sites %>% count(hydrology)
```

```
## # A tibble: 3 x 2
##   hydrology     n
##   <ord>     <int>
## 1 moist       135
## 2 fresh       327
## 3 dry         170
```

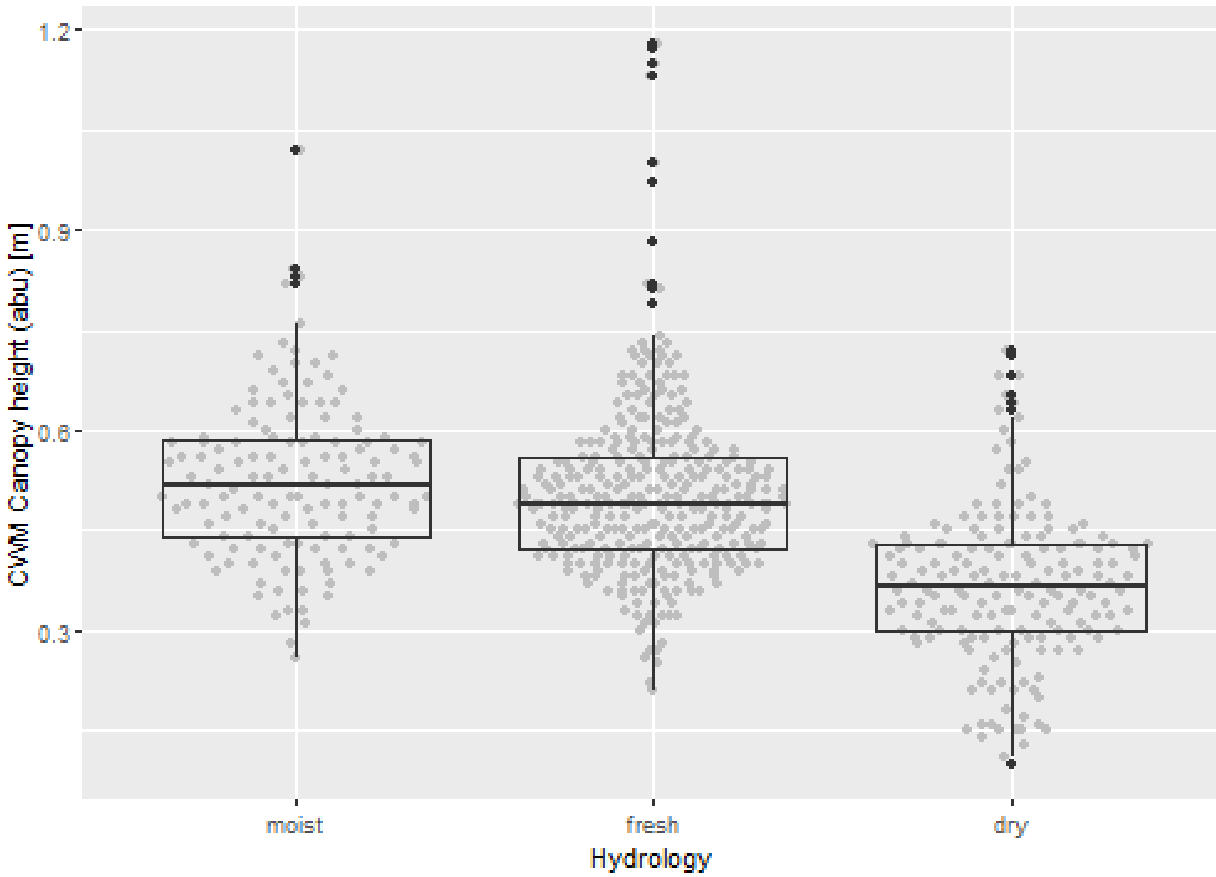
```r
sites %>% count(hydrology, eco.id)
```

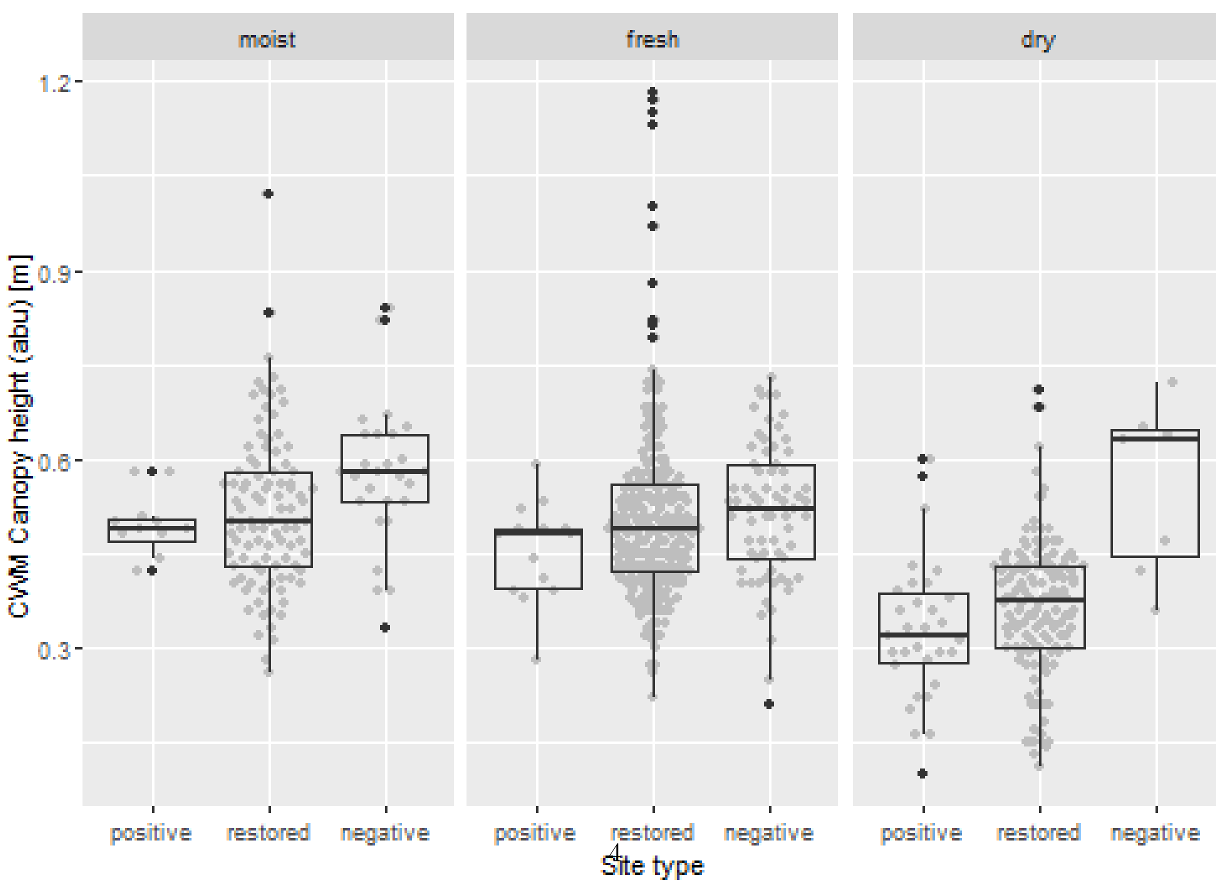
```
## # A tibble: 9 x 3
##   hydrology eco.id     n
##   <ord>     <fct>  <int>
## 1 moist     654       35
## 2 moist     664       50
## 3 moist     686       50
## 4 fresh     654      108
## 5 fresh     664      109
## 6 fresh     686      110
## 7 dry       654       66
## 8 dry       664       53
## 9 dry       686       51
```

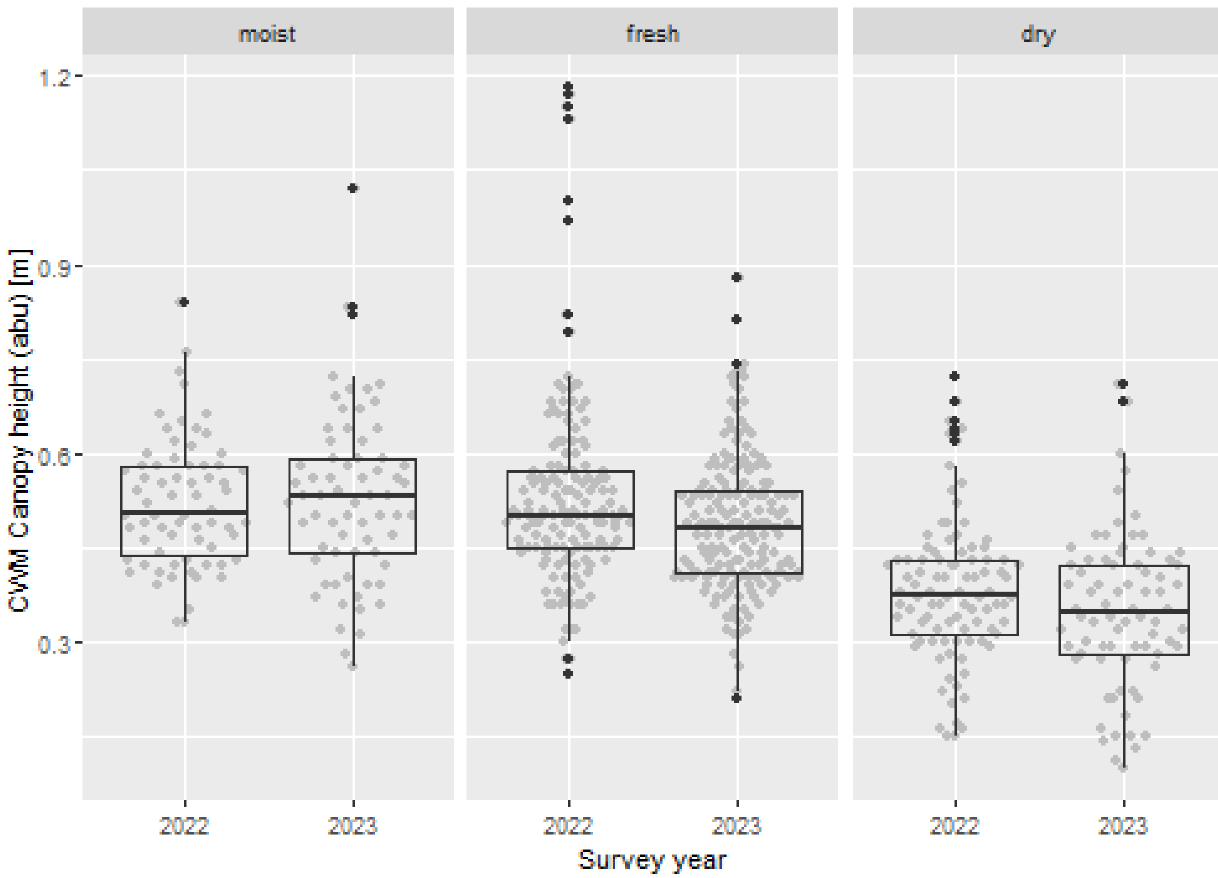
```r
sites %>% count(hydrology, site.type)
```

```
## # A tibble: 9 x 3
##   hydrology site.type     n
##   <ord>     <ord>     <int>
## 1 moist     positive     12
## 2 moist     restored     96
## 3 moist     negative     27
## 4 fresh     positive     14
## 5 fresh     restored    244
## 6 fresh     negative     69
## 7 dry       positive     31
## 8 dry       restored    132
## 9 dry       negative      7
```

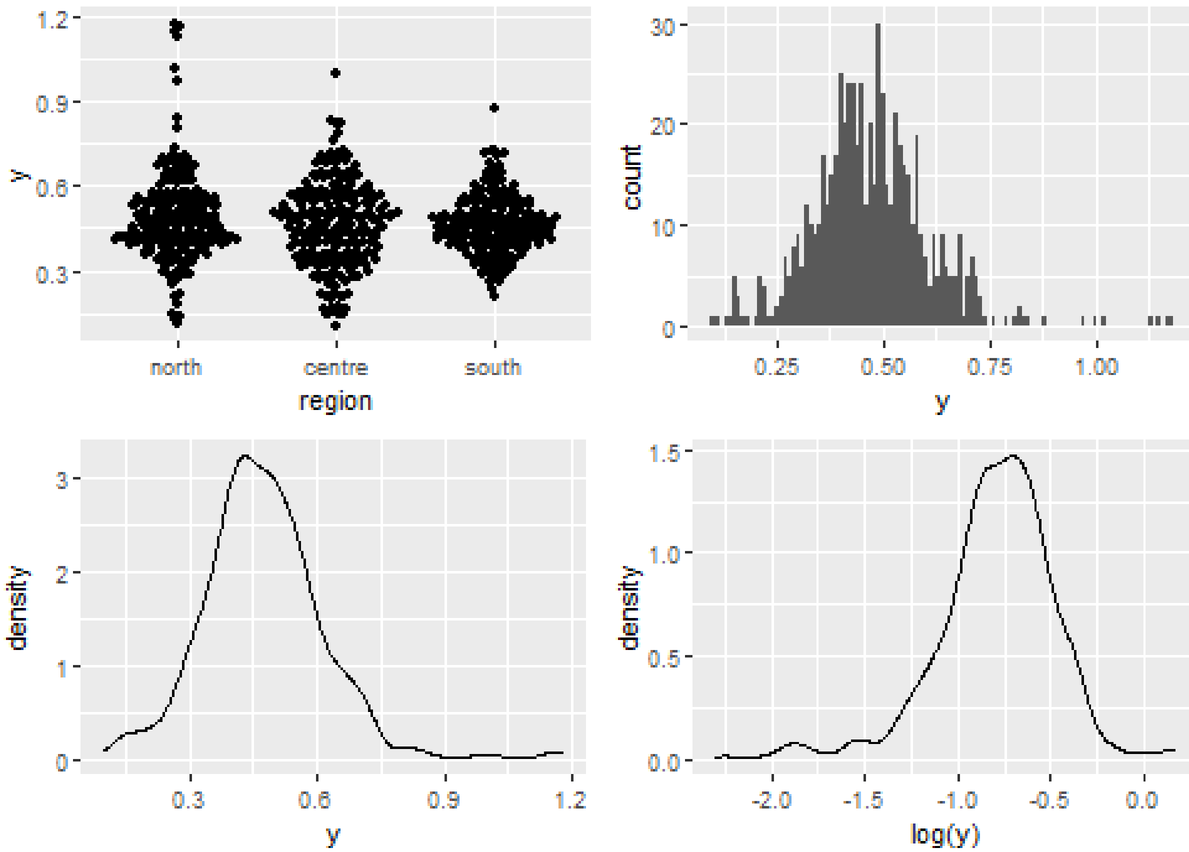
Graphs of raw data (Step 2, 6, 7)






moist
fresh
dry
1.2
0.9
0.6
0.3
CWM Canopy height (abu) [m]
2022
2023
2022
2023
2022
2023
Survey year

Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_height_refs_1.Rdata"))
load(file = here("outputs", "models", "model_height_refs_2.Rdata"))
m_1 <- m1
m_2 <- m2
```

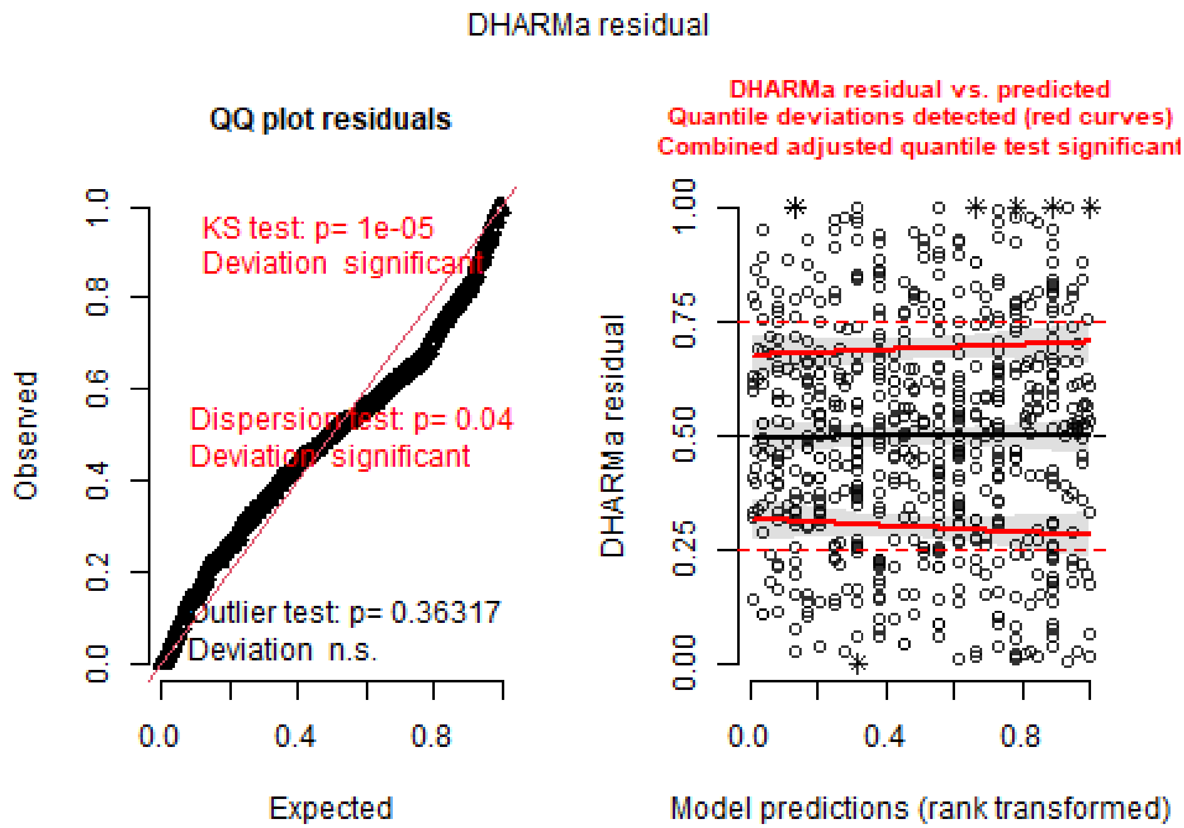
```
m_1@call
## lmer(formula = y ~ hydrology * site.type + eco.id + obs.year +
##     (1 | id.site), data = sites, REML = FALSE)
```

```r
m_2@call
## lmer(formula = y ~ (hydrology + site.type + eco.id + obs.year)^2 +
##     (1 | id.site), data = sites, REML = FALSE)
```

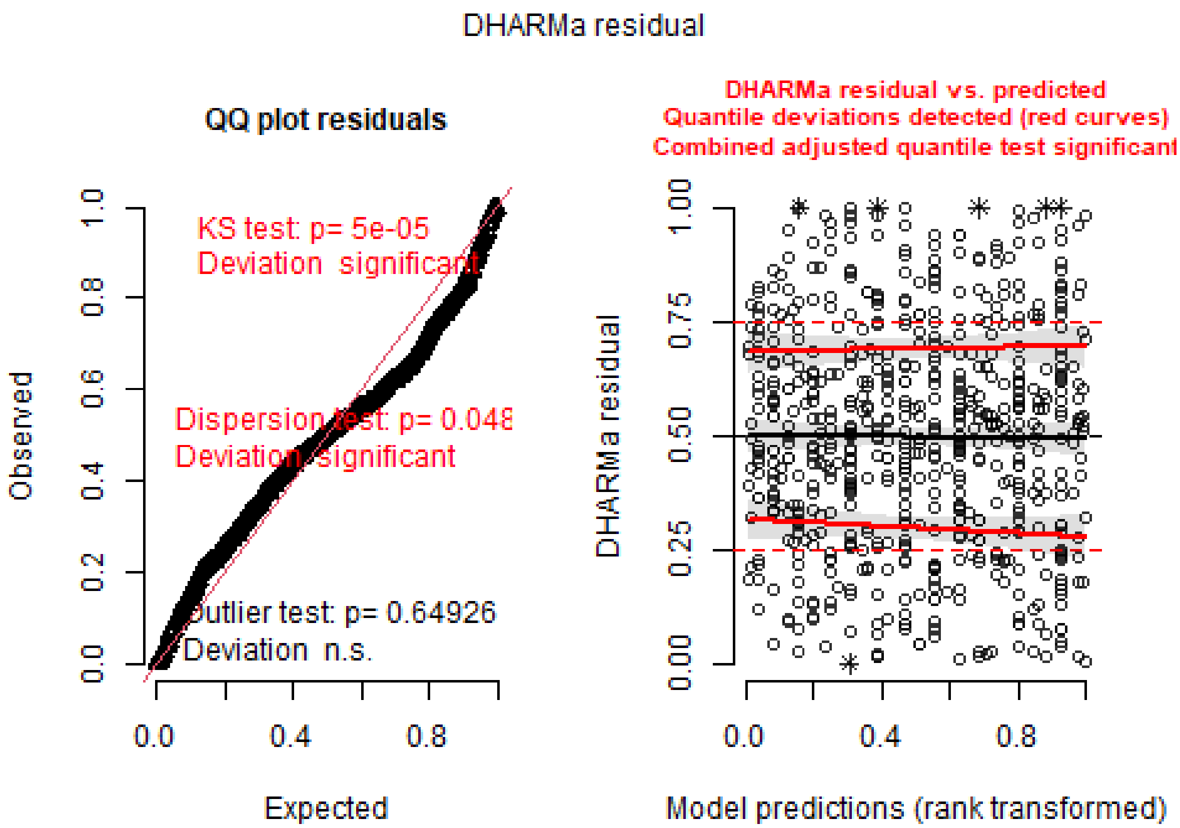
Model check

DHARMa

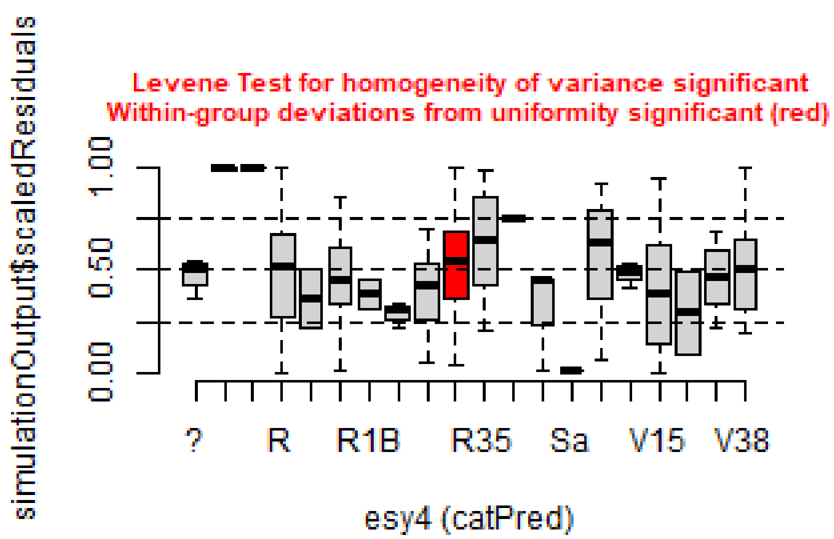
```r
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```



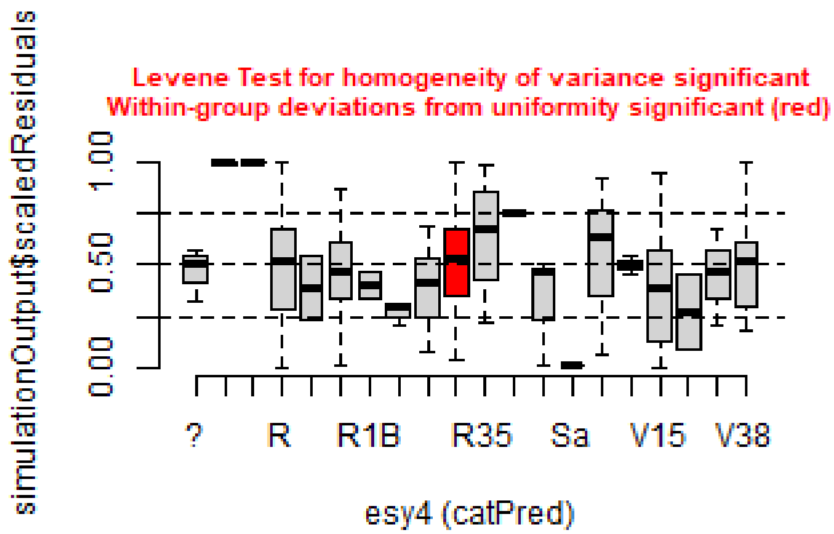

```r
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```



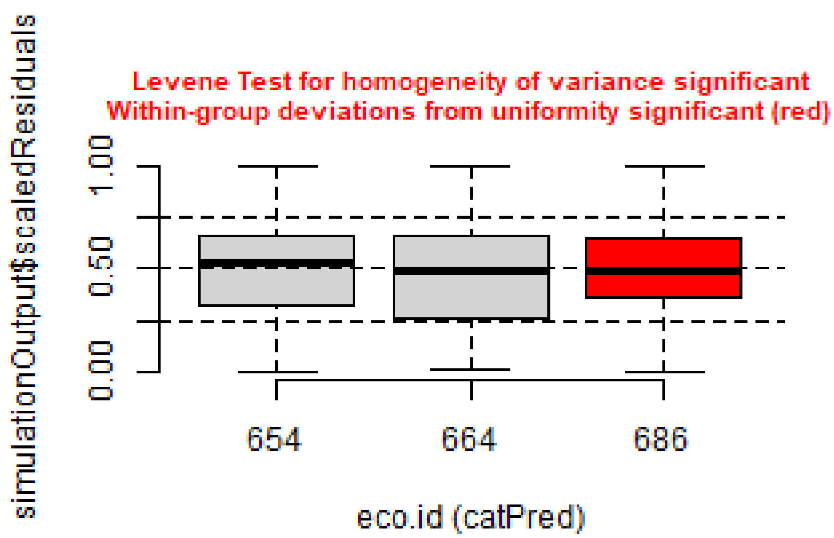

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
## Warning in ensurePredictor(simulationOutput, predictor):
## DHARMa:::ensurePredictor: character string was provided as predictor. DHARMa
## has converted to factor automatically. To remove this warning, please convert
## to factor before attempting to plot with DHARMa.
```



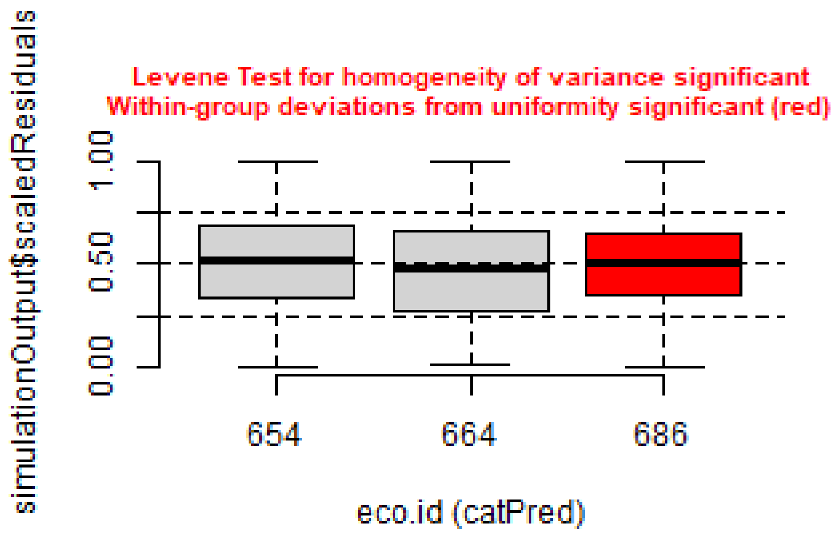
```r
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
## Warning in ensurePredictor(simulationOutput, predictor):
## DHARMa:::ensurePredictor: character string was provided as predictor. DHARMa
## has converted to factor automatically. To remove this warning, please convert
## to factor before attempting to plot with DHARMa.
```



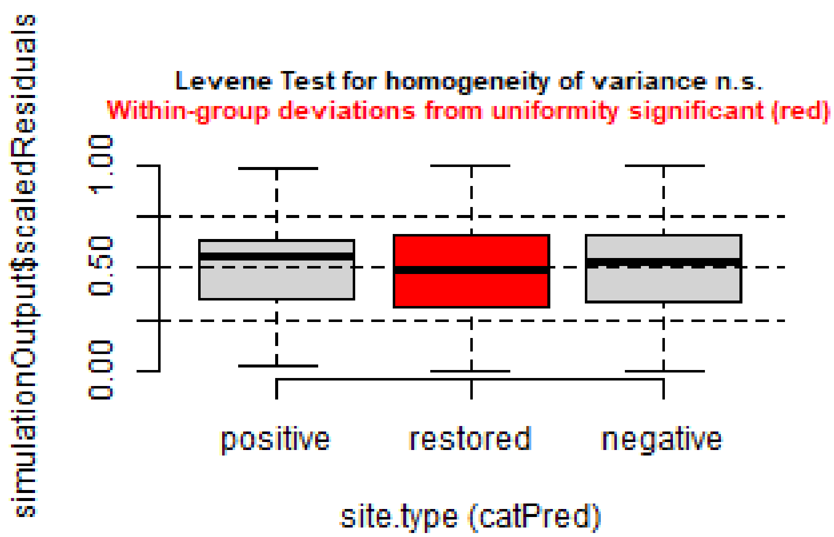

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



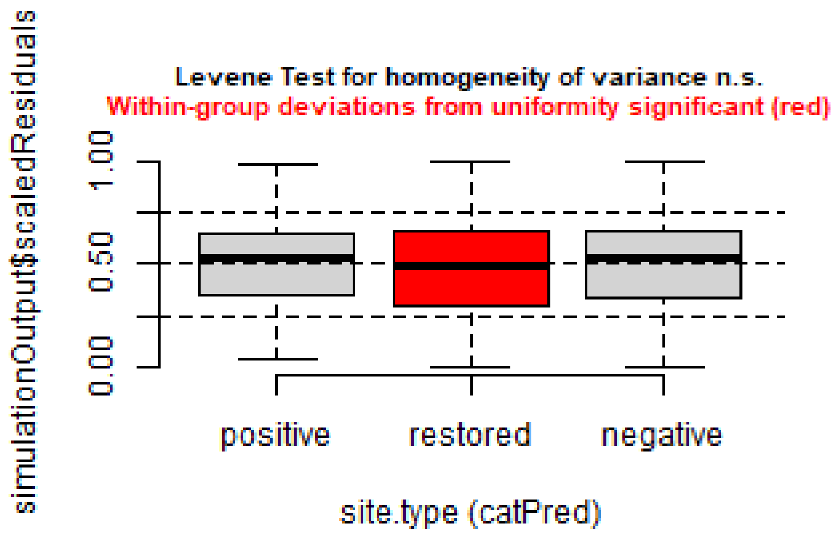

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



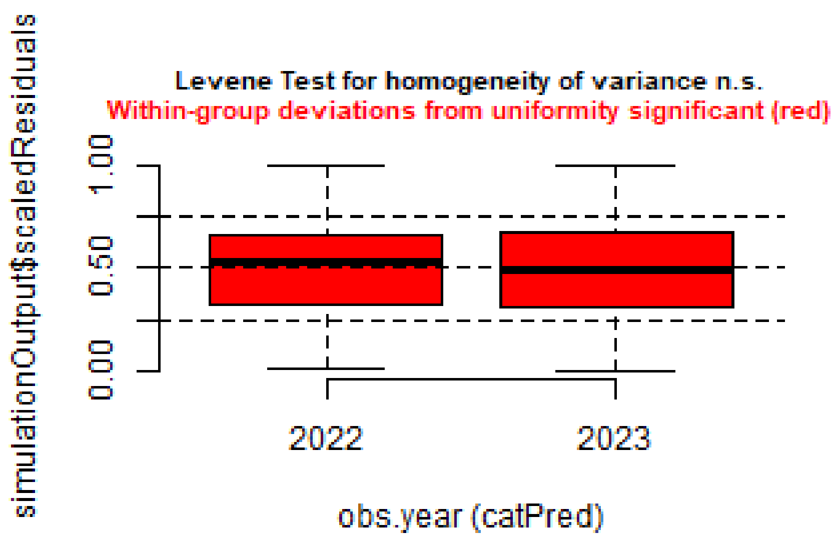

```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



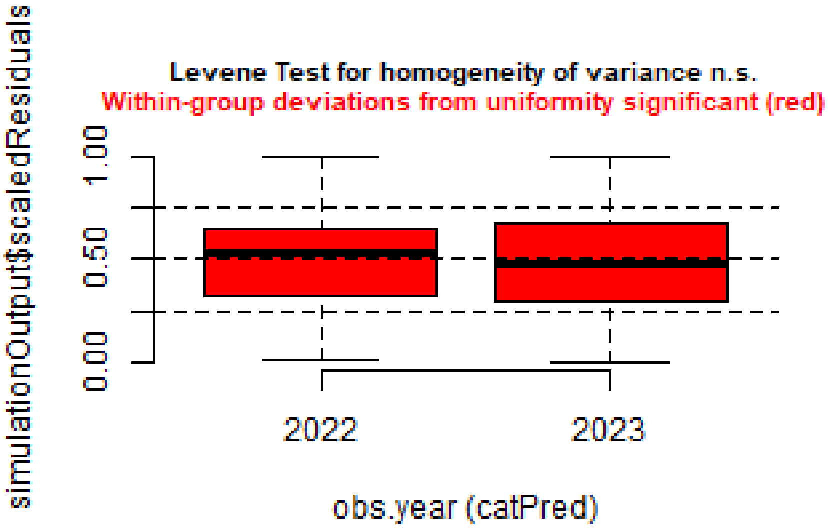

```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



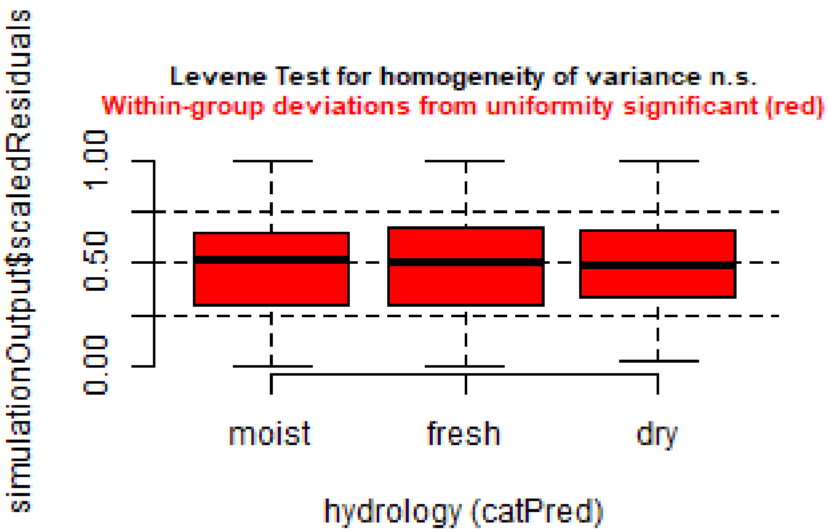

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



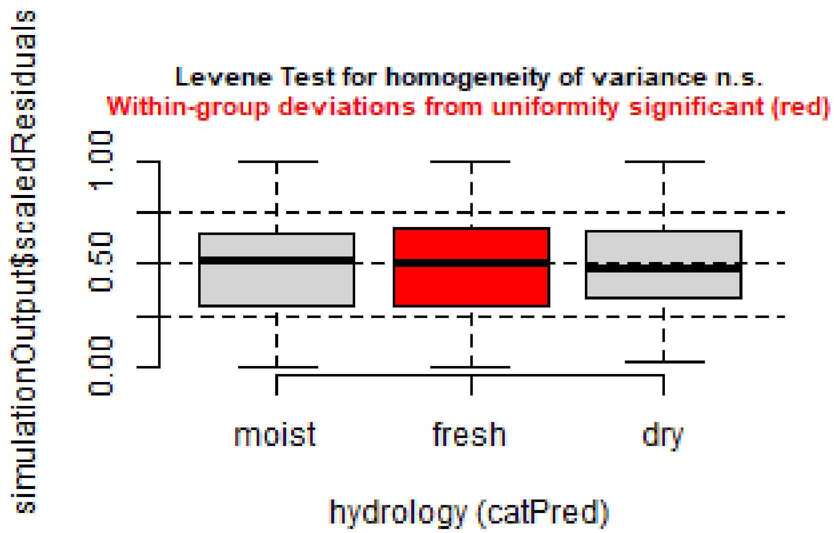

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



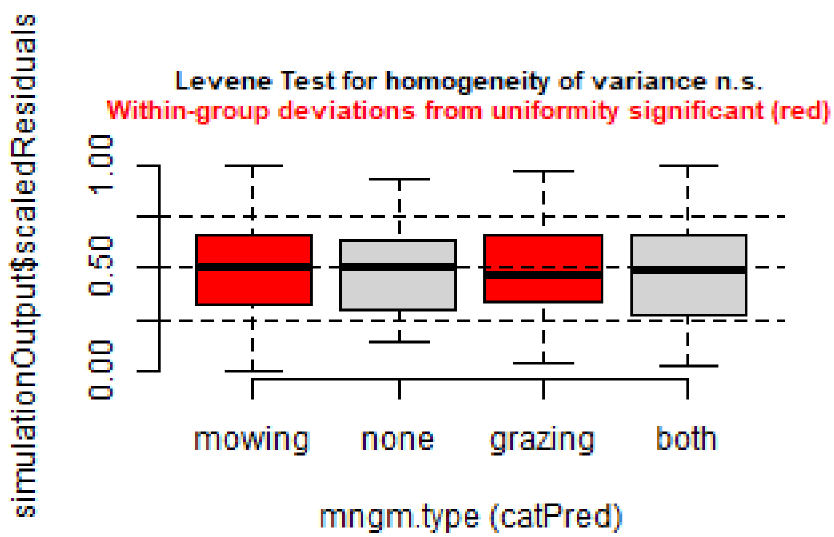

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



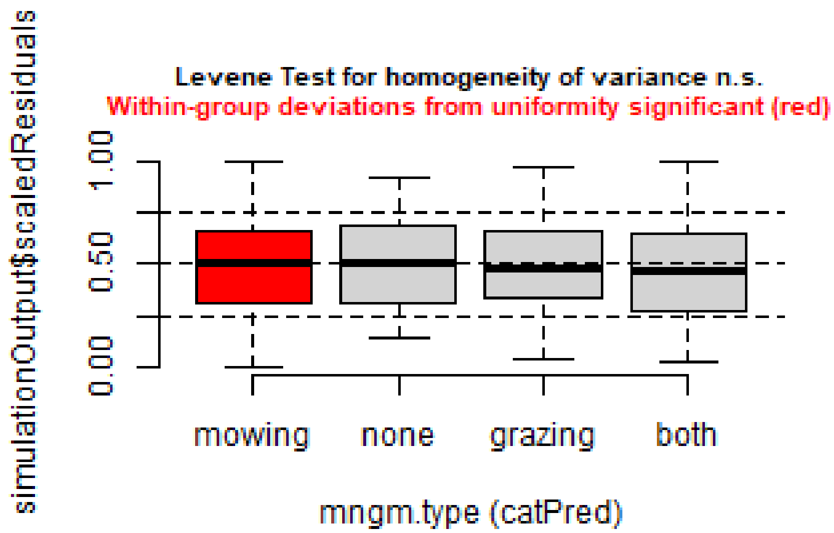

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```




```r
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```




```r
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
## Registered S3 method overwritten by 'car':
##   method           from
##   na.action.merMod lme4
```

```
##                          GVIF Df GVIF^(1/(2*Df))
## hydrology            5.168471  2        1.507789
## site.type            2.214532  2        1.219890
## eco.id               1.038889  2        1.009584
## obs.year             1.059594  1        1.029366
## hydrology:site.type 7.129297  4        1.278294
```

```
car::vif(m_2)
```

```
##                           GVIF Df GVIF^(1/(2*Df))
## hydrology            44.347540  2        2.580580
## site.type            30.568689  2        2.351360
## eco.id                9.727224  2        1.766027
## obs.year              4.370346  1        2.090537
## hydrology:site.type  14.352286  4        1.395131
## hydrology:eco.id     15.862927  4        1.412693
## hydrology:obs.year    5.517613  2        1.532632
## site.type:eco.id     29.500872  4        1.526614
## site.type:obs.year   11.380485  2        1.836708
## eco.id:obs.year       9.103046  2        1.736987
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##            R2m       R2c
```

```r
## [1,] 0.2532449 0.691834
MuMIn::r.squaredGLMM(m_2)
##            R2m       R2c
## [1,] 0.2904578 0.6917012
```

AICc

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```r
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##     df      AICc
## m_1 14 -1080.645
## m_2 28 -1063.663
```

Predicted values

Summary table

```r
car::Anova(m_1, type = 3)
```

```
## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: y
##                        Chisq Df Pr(>Chisq)
## (Intercept)         663.1489  1  < 2.2e-16 ***
## hydrology            10.9924  2   0.004102 **
## site.type             9.8379  2   0.007307 **
## eco.id                1.8811  2   0.390408
## obs.year              3.4869  1   0.061857 .
## hydrology:site.type   5.0154  4   0.285716
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(m_1)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: y ~ hydrology * site.type + eco.id + obs.year + (1 | id.site)
##    Data: sites
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   -1081.3   -1019.0     554.7   -1109.3       618
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.3092 -0.4907 -0.0022  0.4416  4.7386
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  id.site  (Intercept) 0.008811 0.09387
##  Residual             0.006191 0.07868
## Number of obs: 632, groups:  id.site, 172
##
## Fixed effects:
##                            Estimate Std. Error t value
```

```
## (Intercept)              0.49088    0.01906  25.752
## hydrology.L             -0.07856    0.02409  -3.261
## hydrology.Q             -0.02467    0.01977  -1.248
## site.type.L             -0.07919    0.02583  -3.066
## site.type.Q              0.02035    0.01701   1.196
## eco.id664                0.01902    0.01949   0.976
## eco.id686               -0.00667    0.01940  -0.344
## obs.year2023            -0.03022    0.01618  -1.867
## hydrology.L:site.type.L -0.07233    0.04914  -1.472
## hydrology.Q:site.type.L -0.06007    0.04010  -1.498
## hydrology.L:site.type.Q  0.03579    0.03252   1.101
## hydrology.Q:site.type.Q  0.03624    0.02661   1.362
##
## Correlation of Fixed Effects:
##              (Intr) hydr.L hydr.Q st.t.L st.t.Q ec.664 ec.686 o.2023 h.L:..L
## hydrology.L   0.143
## hydrology.Q   0.120  0.202
## site.type.L -0.075 -0.425 -0.397
## site.type.Q  0.486  0.274  0.218 -0.178
## eco.id664   -0.496  0.096 -0.011 -0.072  0.021
## eco.id686   -0.544  0.052  0.038 -0.045 -0.011  0.502
## obs.yer2023 -0.459 -0.033  0.185 -0.034 -0.040 -0.004  0.031
## hydrl.L:..L -0.289 -0.370 -0.327  0.300 -0.383 -0.017  0.059 -0.014
## hydrl.Q:..L -0.297 -0.340  0.089  0.325 -0.366 -0.067  0.029  0.100  0.241
## hydrl.L:..Q  0.205  0.710  0.196 -0.380  0.200  0.042 -0.020 -0.070 -0.338
## hydrl.Q:..Q  0.060  0.205  0.728 -0.362  0.320 -0.018  0.048  0.173 -0.296
##             h.Q:..L h.L:..Q
## hydrology.L
## hydrology.Q
## site.type.L
## site.type.Q
## eco.id664
## eco.id686
## obs.yer2023
## hydrl.L:..L
## hydrl.Q:..L
## hydrl.L:..Q -0.313
## hydrl.Q:..Q  0.083   0.138
```

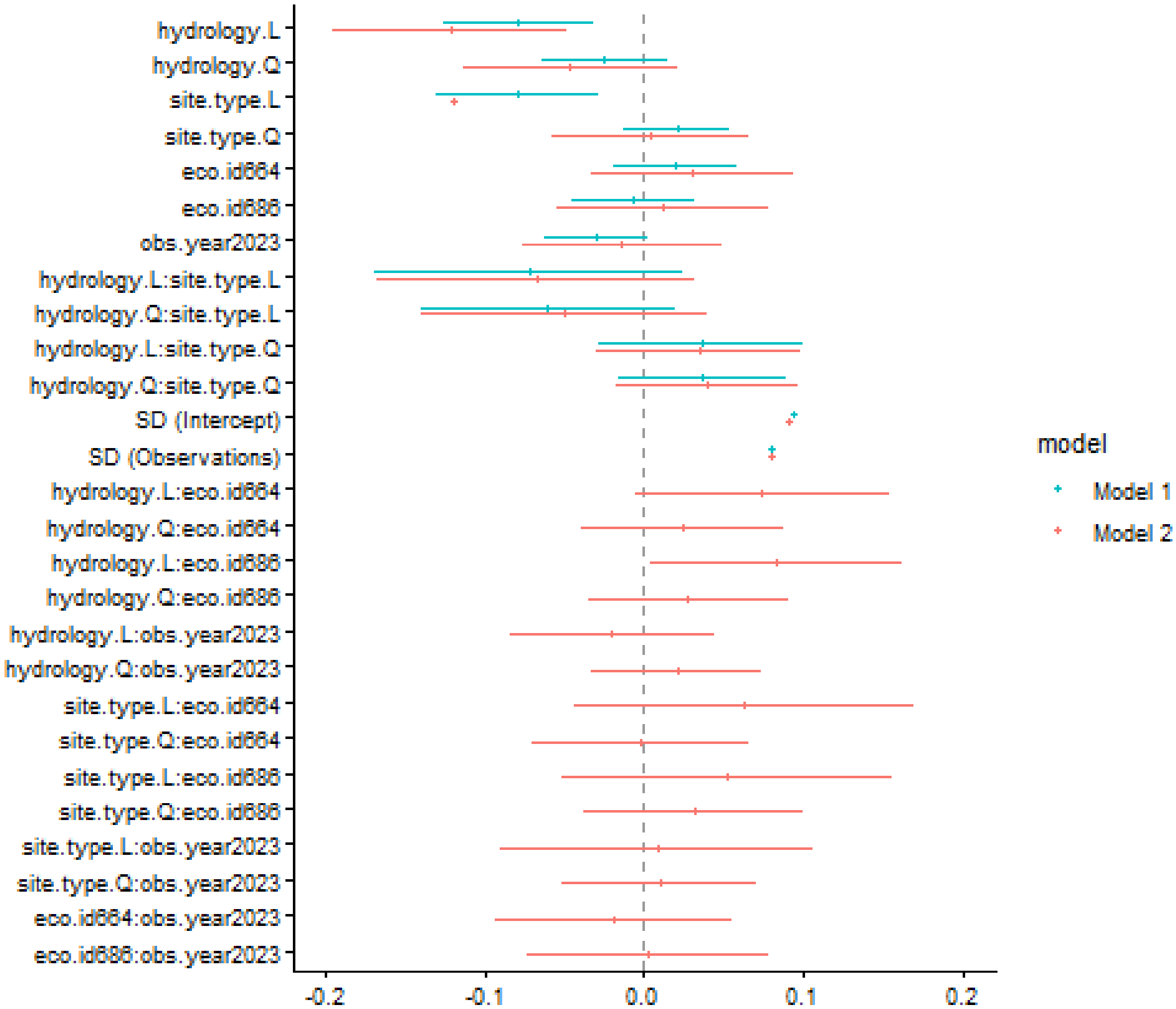
Forest plot

```r
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  xlim(-0.2, 0.2) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
```

```
## Warning: Using the `size` aesthetic with geom_segment was deprecated in ggplot2 3.4.0.
```

```
## i Please use the `linewidth` aesthetic instead.
## i The deprecated feature was likely used in the dotwhisker package.
##   Please report the issue at <https://github.com/fsolt/dotwhisker/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

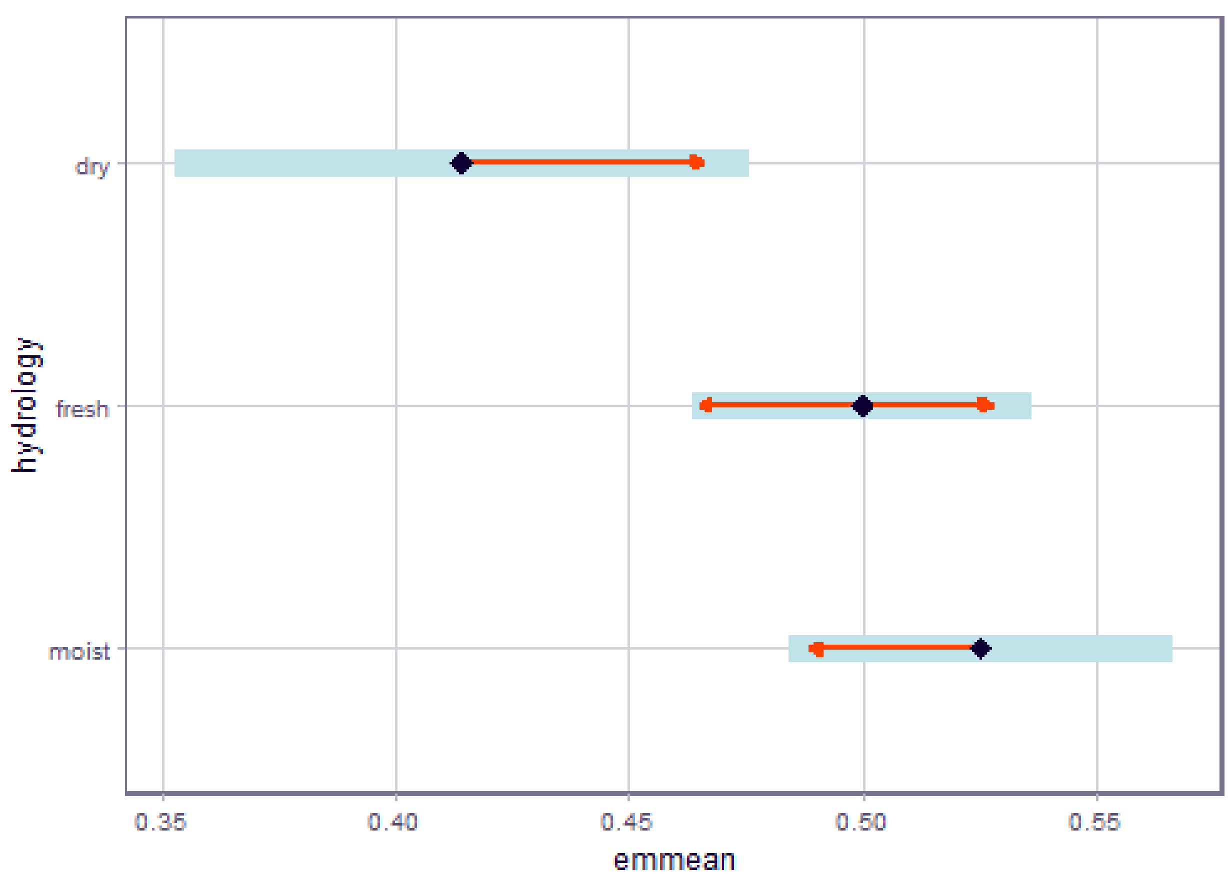
```
(emm <- emmeans(
  m_1,
  revpairwise ~ hydrology,
  type = "response"
  ))
```

Hydrology

```
## $emmeans
```

```
## hydrology emmean     SE    df lower.CL upper.CL
## moist      0.525 0.0208 166.8    0.484    0.566
## fresh      0.500 0.0176  26.5    0.464    0.536
## dry        0.414 0.0284  12.9    0.353    0.476
##
## Results are averaged over the levels of: site.type, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast      estimate     SE   df t.ratio p.value
## fresh - moist  -0.0253 0.0275 61.3  -0.921  0.6291
## dry - moist    -0.1111 0.0352 22.9  -3.153  0.0120
## dry - fresh    -0.0858 0.0334 15.6  -2.567  0.0519
##
## Results are averaged over the levels of: site.type, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
plot(emm, comparison = TRUE)
```



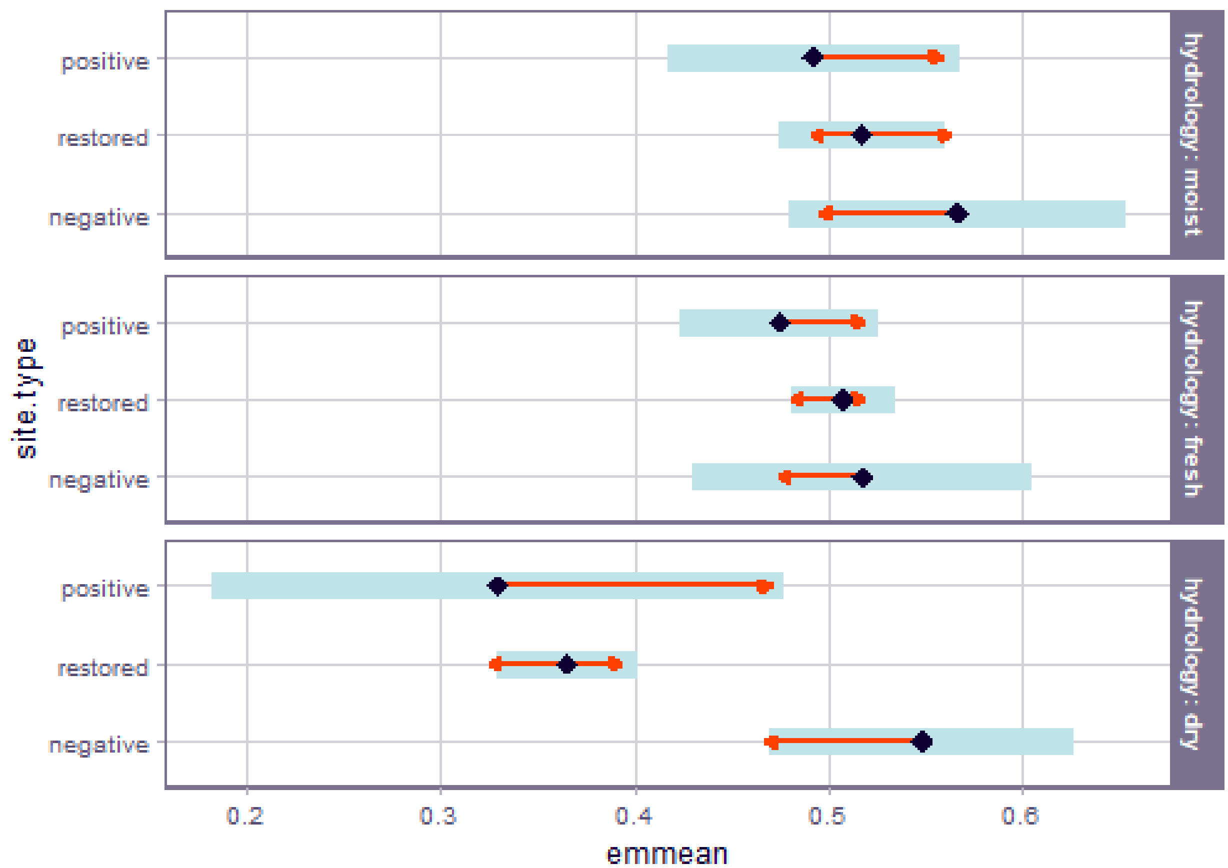

```
(emm <- emmeans(
  m_1,
  revpairwise ~ site.type | hydrology,
```

```
  type = "response"
  ))
```

Hydrology x site type

```
## $emmeans
## hydrology = moist:
##  site.type emmean     SE      df lower.CL upper.CL
##  negative   0.567 0.0445  463.11    0.479    0.654
##  restored   0.517 0.0216  176.95    0.475    0.560
##  positive   0.492 0.0381   97.25    0.417    0.568
##
## hydrology = fresh:
##  site.type emmean     SE      df lower.CL upper.CL
##  negative   0.518 0.0447 2802.00    0.430    0.605
##  restored   0.508 0.0135  177.07    0.481    0.534
##  positive   0.475 0.0241   15.33    0.424    0.526
##
## hydrology = dry:
##  site.type emmean     SE      df lower.CL upper.CL
##  negative   0.548 0.0345    8.31    0.470    0.627
##  restored   0.365 0.0184  177.01    0.329    0.401
##  positive   0.329 0.0756 3928.66    0.181    0.478
##
## Results are averaged over the levels of: eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## hydrology = moist:
##  contrast            estimate     SE      df t.ratio p.value
##  restored - negative -0.04922 0.0494  346.46  -0.996  0.5796
##  positive - negative -0.07434 0.0584  160.73  -1.272  0.4129
##  positive - restored -0.02512 0.0438  108.37  -0.573  0.8348
##
## hydrology = fresh:
##  contrast            estimate     SE      df t.ratio p.value
##  restored - negative -0.00999 0.0467  958.04  -0.214  0.9751
##  positive - negative -0.04263 0.0503   24.38  -0.847  0.6780
##  positive - restored -0.03263 0.0276   17.29  -1.182  0.4792
##
## hydrology = dry:
##  contrast            estimate     SE      df t.ratio p.value
##  restored - negative -0.18355 0.0390    9.08  -4.702  0.0028
##  positive - negative -0.21901 0.0832   11.82  -2.632  0.0537
##  positive - restored -0.03546 0.0778 1189.00  -0.456  0.8918
##
## Results are averaged over the levels of: eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
plot(emm, comparison = TRUE)
```



Session info

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_2.0.3    DHARMa_0.5.0     patchwork_1.3.2  ggbeeswarm_0.7.3
##  [5] lubridate_1.9.5  forcats_1.0.1    stringr_1.6.0    dplyr_1.2.1
##  [9] purrr_1.2.2      readr_2.2.0      tidyr_1.3.2      tibble_3.3.1
```

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## [13] ggplot2_4.0.3    tidyverse_2.0.0  here_1.0.2
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.6           gridExtra_2.3          sandwich_3.1-1
##  [4] rlang_1.2.0            magrittr_2.0.5         multcomp_1.4-30
##  [7] otel_0.2.0             compiler_4.5.2         mgcv_1.9-3
## [10] vctrs_0.7.3            pkgconfig_2.0.3        crayon_1.5.3
## [13] fastmap_1.2.0          backports_1.5.1        labeling_0.4.3
## [16] utf8_1.2.6             ggstance_0.3.7         promises_1.5.0
## [19] rmarkdown_2.31         tzdb_0.5.0             nloptr_2.2.1
## [22] bit_4.6.0              xfun_0.58              later_1.4.8
## [25] broom_1.0.13           parallel_4.5.2         R6_2.6.1
## [28] gap.datasets_0.0.6     stringi_1.8.7          qgam_2.0.0
## [31] RColorBrewer_1.1-3     car_3.1-5              boot_1.3-32
## [34] estimability_1.5.1     Rcpp_1.1.1-1.1         iterators_1.0.14
## [37] knitr_1.51             zoo_1.8-15             parameters_0.29.1
## [40] httpuv_1.6.17          Matrix_1.7-4           splines_4.5.2
## [43] timechange_0.4.0       tidyselect_1.2.1       rstudioapi_0.18.0
## [46] abind_1.4-8            yaml_2.3.12            MuMIn_1.48.19
## [49] doParallel_1.0.17      codetools_0.2-20       lattice_0.22-7
## [52] plyr_1.8.9             bayestestR_0.18.1      shiny_1.13.0
## [55] withr_3.0.2            S7_0.2.2               evaluate_1.0.5
## [58] marginaleffects_0.32.0 survival_3.8-3         pillar_1.11.1
## [61] gap_1.15.2             carData_3.0-6          foreach_1.5.2
## [64] stats4_4.5.2           insight_1.5.1          reformulas_0.4.4
## [67] generics_0.1.4         vroom_1.7.1            rprojroot_2.1.1
## [70] hms_1.1.4              scales_1.4.0           minqa_1.2.8
## [73] xtable_1.8-8           glue_1.8.1             tools_4.5.2
## [76] data.table_1.18.4      lme4_2.0-1             mvtnorm_1.4-0
## [79] grid_4.5.2             rbibutils_2.4.1        datawizard_1.3.1
## [82] nlme_3.1-168           Rmisc_1.5.1            performance_0.17.0
## [85] beeswarm_0.4.0         vipor_0.4.7            Formula_1.2-5
## [88] cli_3.6.6              gtable_0.3.6           digest_0.6.39
## [91] pbkrtest_0.5.5         TH.data_1.1-5          farver_2.1.2
## [94] htmltools_0.5.9        lifecycle_1.0.5        mime_0.13
## [97] bit64_4.8.2            dotwhisker_0.8.4       MASS_7.3-65
```